DAY14-NMAP-BASICS

THEORY (answer all 20 questions in your own words):

1. What does Nmap stand for: Network Mapper

2. What is Nmap used for?

- Used for network discovery, security auditing, and penetration testing.

3. What is network discovery?

- Used to understand host, port, service, network, exposure and system configurations.

4. What is a port scan?

- It is a technique used to probe a device or network to find out which communication ports are open, closed, or filtered, and to identify the active services running on them.

5. What does an open port generally indicate?

- Suitable connections are accepted.

6. Does an open port automatically mean vulnerability? Explain.

- No, an open port does not automatically mean vulnerability, it normally means it listening incoming network traffic on that specific virtual endpoint.

7. What does closed mean conceptually?

- No service accepting connections on port

8. What does filtered mean conceptually?

- Nmap cannot determine network filetering cause firewall, security controls and etc.. prevents normal responses

9. What does nmap 127.0.0.1 do: Scans local machine for open port and services.

10. What does -p do: Specific port will be checked

DAY14-NMAP-BASICS

11. What does -sV do: Gathers service and version

12. What does -oN do: Saves normal readable output to a file

13. Why should scan results be documented?

- To keep track of recordings, findings, identify risk and support future comparisons.

14. What is cybersecurity scope?

- Scope define which system, networks, and devices are authorized for testing.

15. Why must authorization exist before scanning?

- To ensure testing is legal, permitted, and within agreed boundaries.

16. What is the difference between scanning and exploitation?

- Scanning: Collecting network/ service info
- Exploitation: Attempting to use weakness to affect a system.

17. Why is localhost a good beginner target?

- It allows safe practice on your own machine without scanning external systems.

18. How do ports relate to services?

- Ports identify communication endpoints used by network services.

19. Why are unexpected listening services worth reviewing?

- They may indicate unnecessary exposure, misconfiguration, or security risks.

20. Explain a responsible Nmap assessment workflow.

- Obtain permission, define the scope, scan authorised targets, document results, assess risks, and report findings.

DAY14-NMAP-BASICS

DAY 14 PRACTICAL LAB — Your First Authorized Nmap Assessment

Environment: your Kali Linux VM (Nmap is preinstalled on Kali). Target: localhost OR a second VM you personally control/are authorized to use.

TASK 1 — nmap --version → record Nmap Version.

```
┌──(kirthikaa㉿kali)-[~]
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

TASK 2 — nmap --help → what does -p control? (Ports.)

```
┌──(kirthikaa㉿kali)-[~]
└─$ nmap --help
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
```

TASK 3 — nmap 127.0.0.1 → record the number of open ports. ZERO open ports is perfectly acceptable — do NOT start unnecessary services to "get results".

```
┌──(kirthikaa㉿kali)-[~]
└─$ nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-18 13:05 +0800
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000020s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

DAY14-NMAP-BASICS

TASK 4 — nmap -p 22,80,443 127.0.0.1 → build your table: Port / State / Service.

```
┌──(kirthikaa㉿kali)-[~]
└─$ nmap -p 22,80,443, 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-18 13:06 +0800
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00039s latency).

PORT    STATE  SERVICE
22/tcp  closed ssh
80/tcp  closed http
443/tcp closed https

Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
```

TASK 5 — nmap -sV 127.0.0.1 → record any detected service/version; if none: "No service/version detected".

```
┌──(kirthikaa㉿kali)-[~]
└─$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-18 13:06 +0800
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```

TASK 6 — nmap -oN day14-scan.txt 127.0.0.1 then cat day14-scan.txt → confirm your report was saved.

```
┌──(kirthikaa㉿kali)-[~]
└─$ nmap -oN day14-scan.txt 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-18 13:07 +0800
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.22 seconds

┌──(kirthikaa㉿kali)-[~]
└─$ cat day14-scan.txt
# Nmap 7.99 scan initiated Sun Oct 18 13:07:39 2026 as: /usr/lib/nmap/nmap --privileged -oN day14-scan.txt 127.0.0.1
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

# Nmap done at Sun Oct 18 13:07:39 2026 -- 1 IP address (1 host up) scanned in 0.22 seconds
```

TASK 7 (OPTIONAL) — only if you have an authorized second VM: identify its IP, then nmap <AUTHORIZED_LAB_IP>. Skip freely — not required.

TASK 8 — Interpret: 22/tcp open ssh · 80/tcp closed http · 443/tcp filtered https → which is open (22/SSH)? closed (80)? filtered (443)?

- **Port 22:** SSH is listening and accepting connections
- **Port 80:** Host is reachable, but no service listens on the port.
- **Port 443:** Nmap cannot decide whether port is open or not cause it is filtered blocked by firewall, network security or security controls.

DAY14-NMAP-BASICS

TASK 9 — Scenario: a server has TCP 22 open. Does that alone prove vulnerability? (NO.) What would a professional review next? (Is SSH required? Access restricted? Authentication strong? Software maintained? Exposure appropriate?) — never attack instructions.

- No, an open SSH port 22 is not alone enough to prove vulnerability exists.
- Professional would review: Is SSH required?, Is Authentication storng?, Is exposing SSH appropriate for server purpose?.

TASK 10 — Mini report:

TARGET: 127.0.0.1 · AUTHORIZED: YES

OPEN PORTS: Three scanned ports are closed

SERVICES: SSH(22), HTTP(80), HTTPS(443)

OBSERVATIONS: local host is up, but ports are closed.

RECOMMENDATION: Keep unnecessary services diabled only keep necessary one's enabled.